

Text 1

The architecture of the Central Processing Unit (CPU)

The purpose of the CPU

The Central Processing Unit (CPU) is the most important hardware component in a computer. It has two main functions:

- to process data and instructions
- to control the rest of the computer system

All programs and data processing are run in the CPU and all hardware components are, to some extent, controlled by it.

CPU performance

How common characteristics of CPUs affect their performance

Even though today's processors are tremendously fast, their performance can be affected by a number of factors:

- clock speed
- cache size
- number of cores

Clock speed

The CPU contains a clock which is used to coordinate all of the computer's components. The clock sends out a regular electrical pulse which synchronises (keeps in time) all the components.

The frequency of the pulses is known as the clock speed. Clock speed is measured in hertz. The higher the frequency, the more instructions can be performed in any given moment of time.

CPU clocks can sometimes be sped up slightly by the user. This process is known as overclocking. The more pulses per second, the more

fetch-decode-execute cycles that can be performed and the more instructions that are processed in a given space of time. Overclocking can cause long term damage to the CPU as it is working harder and producing more heat.

In the 1980s, processors commonly ran at a rate of between 3 megahertz (MHz) to 5 MHz, which is 3 million to 5 million pulses or cycles per second. Today, processors commonly run at a rate of 3 gigahertz (GHz) to 5 GHz, which is 3 billion to 5 billion pulses or cycles per second.

Cache size

Cache is a small amount of high-speed random access memory (RAM) built directly within the processor. It is used to temporarily hold data and instructions that the processor is likely to reuse.

The bigger its cache, the less time a processor has to wait for instructions to be fetched.

Number of cores

A processing unit within a CPU is known as a core. Each core is capable of fetching, decoding and executing its own instructions.

The more cores a CPU has, the greater the number of instructions it can process in a given space of time. Many modern CPUs are dual (two) or quad (four) core processors. This provides vastly superior processing power.

Text 2

Embedded systems

The purpose and characteristics of embedded systems

As well as general purpose computers, there are other types of computer systems. The most common of these are known as embedded systems.

An embedded system is a small computer that forms part of a larger system, device or machine. Its purpose is to control the device and to allow a user to interact with it. They tend to have one, or a limited number of tasks that they can perform.

Examples of embedded systems include:

- central heating systems
- engine management systems in vehicles
- domestic appliances, such as dishwashers, TVs and digital phones
- digital watches
- electronic calculators
- GPS systems
- fitness trackers

Embedded devices are not usually programmable by a user – the programming is usually done beforehand by the manufacturer. However, it is often possible to upgrade the software on an embedded device. For example, fitness trackers are embedded systems, but the software can often be upgraded by connecting the device to a PC and installing the new software.

Embedded systems can have advantages over general purpose computers in that:

- Their limited number of functions means they are cheaper to design and build.
- They tend to require less power. Some devices run from batteries.
- They do not need much processing power. They can be built using cheaper, less powerful processors.

Text 3

1984

'Is it your opinion, Winston, that the past has real existence?' [...] O'Brien smiled faintly. 'I will put it more precisely. Does the past exist concretely, in space? Is there somewhere or other a place, a world of solid objects, where the past is still happening?'

'No.'

'Then where does the past exist, if at all?'

'In records. It is written down.'

'In records. And — —?'

'In the mind. In human memories.'

'In memory. Very well, then. We, the Party, control all records, and we control all memories. Then we control the past, do we not?'

ORWELL. G, Nineteen Eighty-Four. New York: Signet Classics, 1977